

Hwl Con.

a)

(empty) $\rightarrow n=0$: empty string, $f_0 = 1$

(A,B) $\rightarrow n=1$: A or B in string, $f_1 = 2$

$n=2$: Length 2 strings $\rightarrow (AA, BB, AB, BA)$, $f_2 = 4$

$\rightarrow$ Initial conditions

$$n \geq 3: f_n = f_{n-1} + f_{n-2} + f_{n-3}$$

$S_n = 2^n - f_n$ $\rightarrow$ Recurrence for strings that do

$S_n = f_{n-1} + f_{n-2} + f_{n-3} + 2^{n-3}$ Contain AAA

$$S_n = 2^n - f_{n-1} + f_{n-2} + f_{n-3}$$

$$\text{OR } S_n = 2f_{n-1} - S_{n-4} + 2^{n-4}, \text{ for } n \geq 4$$

$$\hookrightarrow 2f_{n-1} + S_{n-4}$$

$\rightarrow$ see JP example!

$\rightarrow$ Without using complement!

b) For $n < 3$, no string can have AAA since $2^n - f_n$ is the relation.

$\hookrightarrow$ Meaning $S_0 = S_1 = S_2 = 0$

$n=3 \rightarrow 2^3 = 8$ $\rightarrow$ Total strings $\rightarrow$ From: $\{0, 3\}$

Initial conditions: $S_0 = 0, S_1 = 0, S_2 = 0, S_3 = 1$

$S_3 = 1$ b/c only AAA has

$S_4 = 3$ b/c AAAB, BAAA, AABA

have AAA in the

$$n \geq 4: S_n = 2^n - f_n, f_n = f_{n-1} + f_{n-2} + f_{n-3}$$

c) $n=7, f_n$ For $n: \{0, 7\}$, $f_0 = 1, f_1 = 2, f_2 = 4, 2^7 = 128$ $\rightarrow$ Total length of strings

For $n \geq 3: f_n = f_{n-1} + f_{n-2} + f_{n-3}$, use $2^n - f_n$ or $f_n + 2^{n-3}$

$$f_3 = f_2 + f_1 + f_0 = 4 + 2 + 1 = 7$$

$$f_4 = f_3 + f_2 + f_1 = 7 + 4 + 2 = 13$$

OR use $f_n + 2^{n-3}$

$$f_5 = f_4 + f_3 + f_2 = 13 + 7 + 4 = 24$$

$$f_6 = f_5 + f_4 + f_3 = 24 + 13 + 7 = 44$$

$$f_7 = f_6 + f_5 + f_4 = 44 + 24 + 13 = 81$$

$$S_7 = 128 - 81 = 47$$

Final ans.: 47